

Assignment-5: MCMC

Markov Chain Monte Carlo

January 19

IMPORTANT INSTRUCTIONS

- All questions in this assignment are coding questions.
- **SUBMISSION FORMAT:**
 - Create one Notebook (.ipynb) file
 - Solve all three questions in this single notebook
 - Add clear markdown cells with question titles
 - Include comments in your code to explain it wherever needed
 - Upload the completed (.ipynb) file to Google Classroom

For queries contact us on Whatsapp.

Question 1: Metropolis Algorithm

Implement the Metropolis algorithm to sample from a Gaussian distribution with mean $\mu = 5$ and standard deviation $\sigma = 1$.

(a) Write code to define the target distribution $f(x) = \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ and implement the Metropolis sampler using a Gaussian random walk proposal $x' = x_t + \epsilon$ where $\epsilon \sim \mathcal{N}(0, 1)$. Run for 5000 iterations starting from $x_0 = 0$. Report the empirical mean and standard deviation after 20% burn-in and compare with true values.

(b) Create two plots: a trace plot showing the evolution of the chain over all iterations (mark burn-in period with a vertical line), and a histogram of samples after burn-in overlaid with the true Gaussian distribution.

Question 2: Metropolis-Hastings with Asymmetric Proposal

Implement the Metropolis-Hastings algorithm to sample from an exponential distribution with rate $\lambda = 0.5$.

(a) Define the exponential target $f(x) = 0.5e^{-0.5x}$ for $x > 0$. Use an asymmetric proposal where $x' = x_t + Y$ with $Y \sim \text{Exponential}(1)$. Calculate the acceptance probability as $\alpha = \min\left(1, \frac{f(x')}{f(x_t)} \cdot \frac{q(x_t|x')}{q(x'|x_t)}\right)$. Run for 5000 iterations starting from $x_0 = 1$ and report empirical mean (should be close to $1/\lambda = 2$).

(b) Plot the trace plot and histogram after burn-in. Explain in a brief comment why we need the proposal ratio correction in Metropolis-Hastings.

Question 3: Coin Bias Inference

Use Metropolis algorithm to infer coin bias from observed coin flips.

(a) Generate synthetic data by flipping a coin 100 times with true bias $p_{true} = 0.6$. Define the posterior as $f(p) \propto p^{n_h}(1-p)^{n_t}$ where n_h is number of heads and n_t is number of tails. Run Metropolis for 10000 iterations with proposal std = 0.1,

ensuring $p \in [0, 1]$. Report posterior mean and 95% credible interval.

(b) Create a histogram of posterior samples after burn-in. Compare the posterior mean and credible interval with the true value $p = 0.6$ and explain whether the algorithm recovered the true bias.

HINTS

Question 1:

- Use `np.exp()` and `np.random.normal()`
- Acceptance rate should be around 40-50% for 1D

Question 2:

- Always include proposal ratio correction for asymmetric proposals
- Check that proposals stay positive: reject if $x' \leq 0$
- For exponential: $q(a|b)/q(b|a)$ depends on direction of movement

Question 3:

- Generate heads: `heads = np.random.binomial(100, 0.6)`
- Tails: `tails = 100 - heads`
- Use log to avoid overflow: `log_posterior = heads * np.log(p) + tails * np.log(1-p)`